INDIAN INSTITUTE OF TECHNOLOGY, DELHI

ELL1000 - Project Report

Pulse Width Modulation (PWM) Based Speed Control of a DC Motor using a 555 Timer

Submitted by-

Student Name: Spraha Vats
Entry Number: 2025CE12028

Student Name: Veeraj Jain
Entry Number: 2025CE11853

Objective/Aim

To design and implement a Pulse Width Modulation (PWM) circuit using an NE555 timer in an astable multivibrator configuration to control the angular velocity of a permanent magnet DC motor. The project demonstrates the efficiency of power control via duty cycle modulation compared to linear voltage regulation.

Theoretical Foundation

1. **Pulse Width Modulation (PWM):** Instead of varying the analog voltage level (which wastes energy as heat), we switch the motor "ON" and "OFF" at a high frequency. The average voltage (V_{avg}) delivered to the motor is proportional to the **Duty Cycle (D)**:

$$V_{avg} = V_{source} \times D = V_{source} \times \left(\frac{T_{on}}{T_{on} + T_{off}} \right)$$

2. **Astable Multivibrator:** The 555 Timer is configured to oscillate continuously. By using a potentiometer and diodes, we create separate paths for charging and discharging the timing capacitor, allowing us to vary the duty cycle while keeping the frequency relatively constant.

3. **Transistor Switching:** Since the 555 IC cannot provide enough current to drive a motor directly, a **TIP122 Darlington Pair** acts as a current amplifier/switch, controlled by the PWM signal from Pin 3.

Components

Component	Description
555 Timer IC	A highly versatile integrated circuit used to generate precise timing pulses or oscillations. In this circuit, it functions as the "brain" to create the PWM signal.o
TIP122 NPN Transistor	A Darlington power transistor that acts as a high-current switch. It allows the low-power signal from the 555 timer to control the high-current DC motor safely.
500K Potentiometer	A three-terminal variable resistor is used here to adjust the charging and discharging time of the capacitor, effectively changing the PWM duty cycle and motor speed.
100µF Capacitor	An electrolytic capacitor used to stabilize the power supply or set the timing interval of the oscillator by storing and releasing electrical charge at a specific rate.
1N4148/4007 Diodes	These steering diodes create separate paths for current during the charge/discharge cycles of the potentiometer, enabling duty cycle adjustment from near 0% to 100%.

DC Motor & Propeller	The output transducer that converts electrical energy into mechanical energy. The propeller provides a visual and physical load to demonstrate the changes in rotational speed.
LED	A semiconductor light source used as a visual indicator. Its brightness fluctuates with the PWM signal, providing a secondary confirmation of the duty cycle changes.
1k & 220Ω Resistors	Passive components used to limit current. The 1k protects the transistor base, while the 220Ω prevents the LED from burning out by restricting current flow.
Breadboard	A reusable solderless platform with internal conductive strips, used for prototyping the circuit and making temporary electrical connections between components during the testing phase.

Base experiments

Network Foundations (Exp 2): The circuit relies on **KVL and KCL** to manage current distribution between the timing network and the power stage. **Thevenin's Theorem** is applied when calculating the voltage divider levels inside the 555 IC.

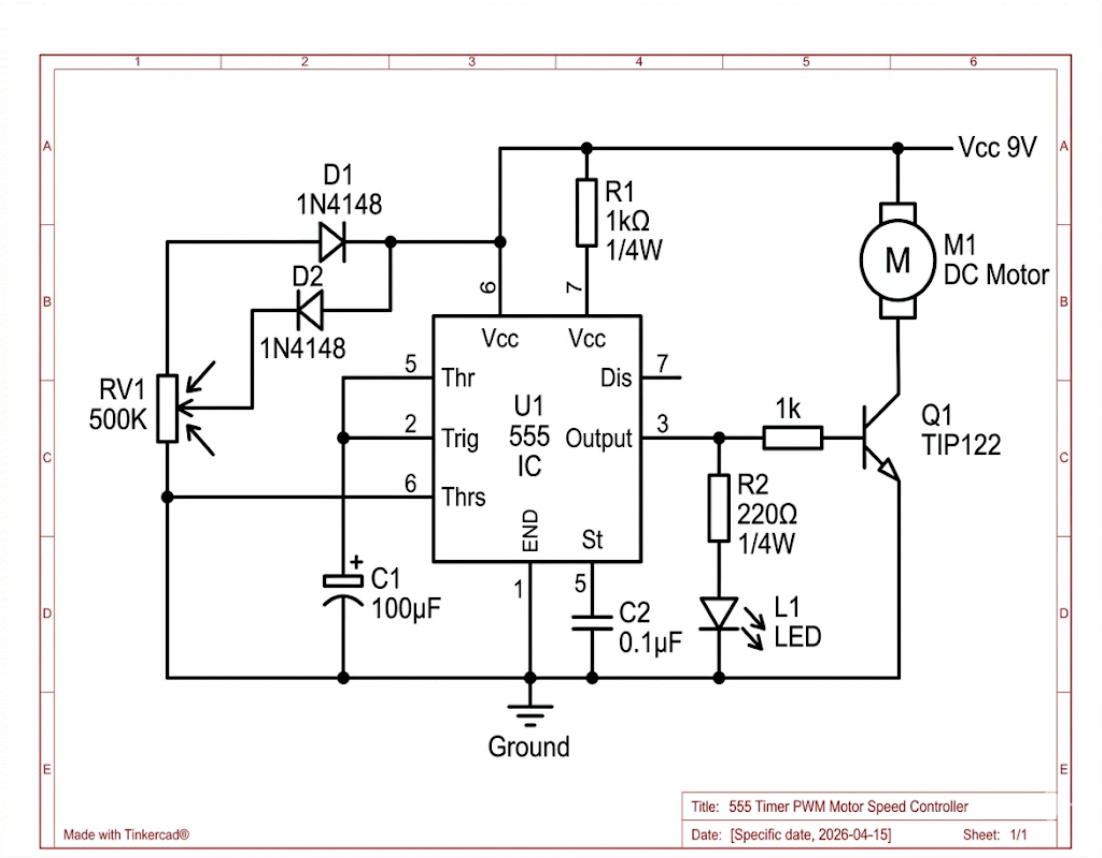
Filter Characteristics (Exp 4): The DC motor behaves as a **Low-Pass Filter**. Due to its internal inductance (L) and mechanical inertia, it "filters out" the high-frequency square wave of the PWM and responds only to the average DC voltage, resulting in smooth speed control rather than jittery motion.

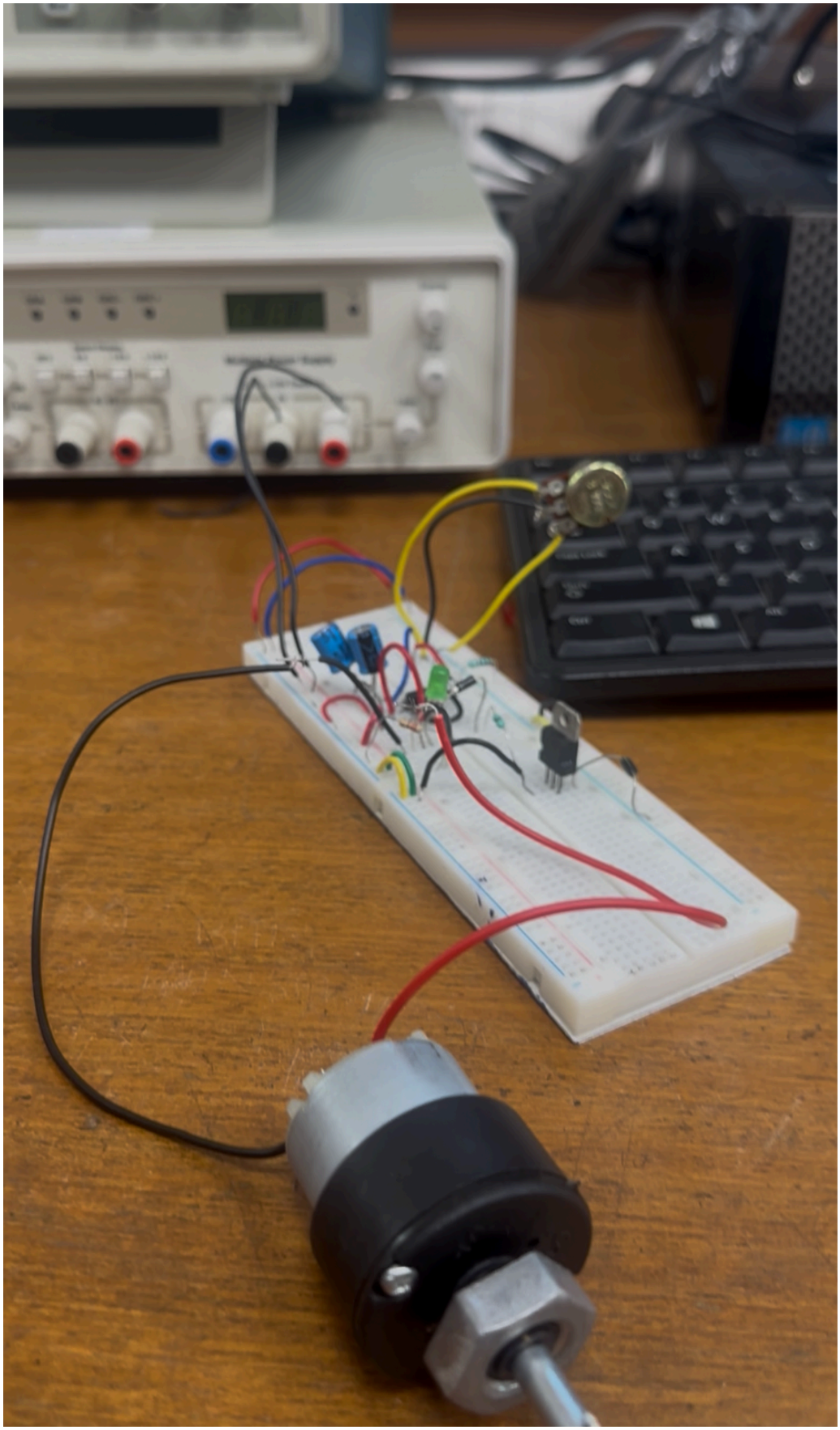
Non-Linear Electronics (Exp 5): The **1N4148/4007 diodes** are used for "steering" the charging and discharging currents to allow independent control of T_{on} and T_{off}. Additionally, a diode acts as a **Flyback Diode** to protect the TIP122 from inductive voltage spikes.

Transient Analysis (Exp 6): The core timing is a direct application of **RC Step Response**. The 555 timer monitors the exponential charging and discharging of the **100µF capacitor**; the time it takes to reach specific voltage thresholds determines the PWM frequency.

Integrated Circuitry (Exp 7): While Exp 7 focuses on Op-Amps, the 555 IC is a practical extension. It uses **internal comparators** (similar to Op-Amps) to trigger the internal flip-flop, transitioning from theoretical signal amplification to practical **Pulse Width Modulation**.

Circuit diagram





Working of the Circuit

The circuit operates as an **Astable Multivibrator** with a modified duty cycle.

1. **Timing Phase:** The 100µF capacitor (C1) charges through one half of the **500k Potentiometer** and one **diode**. When the voltage reaches 2/3 V_{cc}, the 555 internal flip-flop resets.

2. **Discharge Phase:** C1 then discharges through the other **diode** and the remaining half of the potentiometer. When the voltage drops to 1/3 V_{cc}, the flip-flop sets, and the cycle repeats.

3. **Speed Control:** By turning the potentiometer, you change the ratio of charge time (T_{on}) to discharge time (T_{off}). This creates a **PWM signal** at Pin 3.

4. **Power Stage:** The low-power PWM signal cannot drive the motor directly. It triggers the **TIP122 Darlington Transistor**, which acts as a heavy-duty switch, connecting the motor to ground in rapid pulses. The motor's inertia smooths these pulses into a continuous rotation speed.

Applications

- **Industrial Automation:** Controlling the speed of conveyor belts or small robotic actuators.

- **Consumer Electronics:** Adjusting fan speeds in PCs or regulating brightness in LED dimmers.

- **Electric Vehicles:** Basic throttle control for small scale RC cars or drones.

Advantages

- **High Efficiency:** Unlike variable resistors, PWM switching wastes very little power as heat.
- **Full Torque:** The motor receives full voltage pulses even at low speeds, preventing stalling.
- **Cost-Effective:** Uses standard, inexpensive components instead of complex microcontrollers.

Limitations

- **EMI/Noise:** Rapid switching can create electromagnetic interference with nearby sensitive electronics.
- **No Feedback:** The circuit is "open-loop"; it cannot auto-adjust speed if a heavy load is applied to the propeller.
- **Frequency Limits:** At very low frequencies, the motor may vibrate or "stutter."

Future Scope

- **Closed-Loop Control:** Adding an optical encoder or tachometer to keep RPM constant under load.
- **Microcontroller Integration:** Replacing the 555 timer with an **Arduino** (as seen in your format sample) for programmable speed profiles.
- **Regenerative Braking:** Modifying the H-bridge stage to recover energy when the motor slows down.

Conclusion

The experiment successfully demonstrates that speed control of a DC motor can be achieved efficiently using Pulse Width Modulation. By integrating concepts of **RC time constants (Exp 6)** and **Transistor switching**, we established a method to regulate mechanical output via a high-frequency electronic signal.

Possible Errors

- **Component Tolerance:** The 100µF capacitor might have a ±20% tolerance, shifting the expected PWM frequency.
- **Parallax Error:** Inaccuracies in reading the potentiometer position or RPM if measured manually.
- **Heat Dissipation:** If the TIP122 is not switched fully "ON," it can enter the linear region and overheat, dropping the output voltage.

Precautions

- **Polarity Check:** Ensure the 100µF electrolytic capacitor and the TIP122 are oriented correctly to prevent component failure.
- **Flyback Protection:** While your list had 2 diodes, normally a third is used across the motor to prevent high-voltage spikes from damaging the TIP122.
- **Common Ground:** Ensure the power supply for the motor and the 555 IC share a common ground point.

References

1. *Sedra, A. S., & Smith, K. C.* "Microelectronic Circuits," Oxford University Press.
2. *Boylestad, R. L.* "Electronic Devices and Circuit Theory," Pearson.
3. *TI NE555 Datasheet:* "Precision Timers," Texas Instruments.